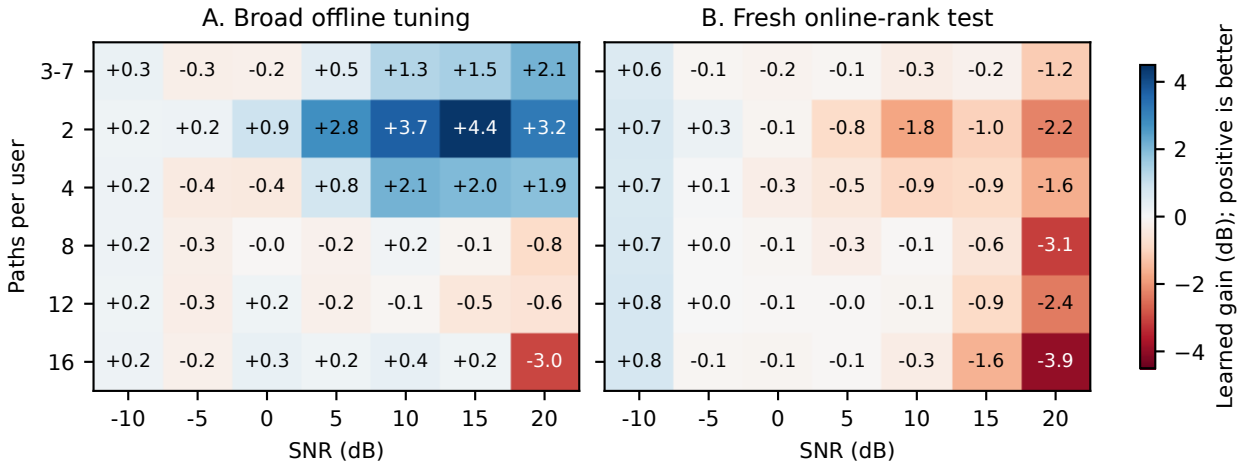
A. Broad offline tuning
B. Fresh online-rank test
Paths per user
3-7
2
4
8
12
16
+0.3 -0.3 -0.2 +0.5 +1.3 +1.5 +2.1
+0.2 +0.2 +0.9 +2.8 +3.7 +4.4 +3.2
+0.2 -0.4 -0.4 +0.8 +2.1 +2.0 +1.9
+0.2 -0.3 -0.0 -0.2 +0.2 -0.1 -0.8
+0.2 -0.3 +0.2 -0.2 -0.1 -0.5 -0.6
+0.2 -0.2 +0.3 +0.2 +0.4 +0.2 -3.0
+0.6 -0.1 -0.2 -0.1 -0.3 -0.2 -1.2
+0.7 +0.3 -0.1 -0.8 -1.8 -1.0 -2.2
+0.7 +0.1 -0.3 -0.5 -0.9 -0.9 -1.6
+0.7 +0.0 -0.1 -0.3 -0.1 -0.6 -3.1
+0.8 +0.0 -0.1 -0.0 -0.1 -0.9 -2.4
+0.8 -0.1 -0.1 -0.1 -0.3 -1.6 -3.9
-10 -5 0 5 10 15 20
SNR (dB)
-10 -5 0 5 10 15 20
SNR (dB)
Learned gain (dB); positive is better
4
2
0
−2
−4